

CHAPTER 22

PERIODIZATION

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After completing this chapter, you will be able to

- understand the central concepts that underpin the periodization of training;
- appreciate the value, role, and application of periodization in strength and conditioning programs;
- describe the four periods of the traditional periodization model;
- describe the two phases of the preparatory period of the traditional periodization model;
- relate the four sport seasons to the four periods of the traditional periodization model; and
- apply the program design variables to create a periodized strength training program.

The ability to guide the training process and implement programming strategies that enhance the athlete's potential for achieving the highest level of performance at targeted time points is largely affected by the long- and short-term programming strategies implemented by the strength and conditioning professional (69). The process of organizing these programming strategies is generally referred to as **periodization**. The basic concepts associated with periodization can be traced to ancient philosophers and physicians, such as Galen (also known as Claudius Aelius Galenus), who presented the first description of the application of periodized strength training methods, and Lucius Flavius Philostratus, who presented a structured pre-Olympic training model (28, 50). In the modern Olympic era periodization is often attributed to Leonid Matveyev (64), who in the 1960s proposed the basic theories that underpin periodization. Even though Matveyev is often considered the father of periodization, it is important to

note that several other individuals were also exploring these concepts at the same time, including László Nádori (71), Tudor Bompa (3), and Yuri Verkhoshansky (102). Later on, American sport scientists Michael H. Stone, Harold O'Bryant, and John Garhammer adapted the concepts of the early periodization theorists with special application to strength and power athletes (89, 91).

While the concepts that underpin our understanding of periodization are well established and are well accepted, confusion still exists in both the scientific (1, 2, 54, 79) and practical (25) literature about what periodization is and how it is used to guide the training process. At the center of this confusion is the fact that the terms *periodization* and *programming* are often used interchangeably within the literature (20, 35, 37, 88). While periodization and programming are interrelated, they represent different management levels of the training process (35, 38). Specifically, periodization serves as the macro-management level of the training process, while programming allows for the micro-management of the process (38). Conceptually, periodization serves as a method for organizing deliberate practice by providing the scaffold from which programming decisions are made (60). The objectives targeted by the periodized plan are then aligned with training cycles (e.g., micro-, meso-, and macrocycles, and annual and multiyear cycles) (73) and individualized programs are constructed. Programming aligns the activities undertaken (i.e., modes and methods of training) as part of the training process with the athlete's established training goals and the direction of the training process (41). At the programming level, an integrated monitoring program is used to fine-tune the training activities by providing information that can further inform decisions about the training strategies implemented as part of the athlete's training program ([figure 22.1](#)) (22). In this regard, programming becomes an operation in strategic thinking, with programmatic decisions being considered in the context of the athlete's holistic training program (e.g., physical training, technical and tactical training, mental skills training, recovery methods, and nutritional strategies) and the current state (37, 77). Ultimately, when all of this is taken into account periodization simply serves as an organizational tool for guiding the direction and structure of the training process.

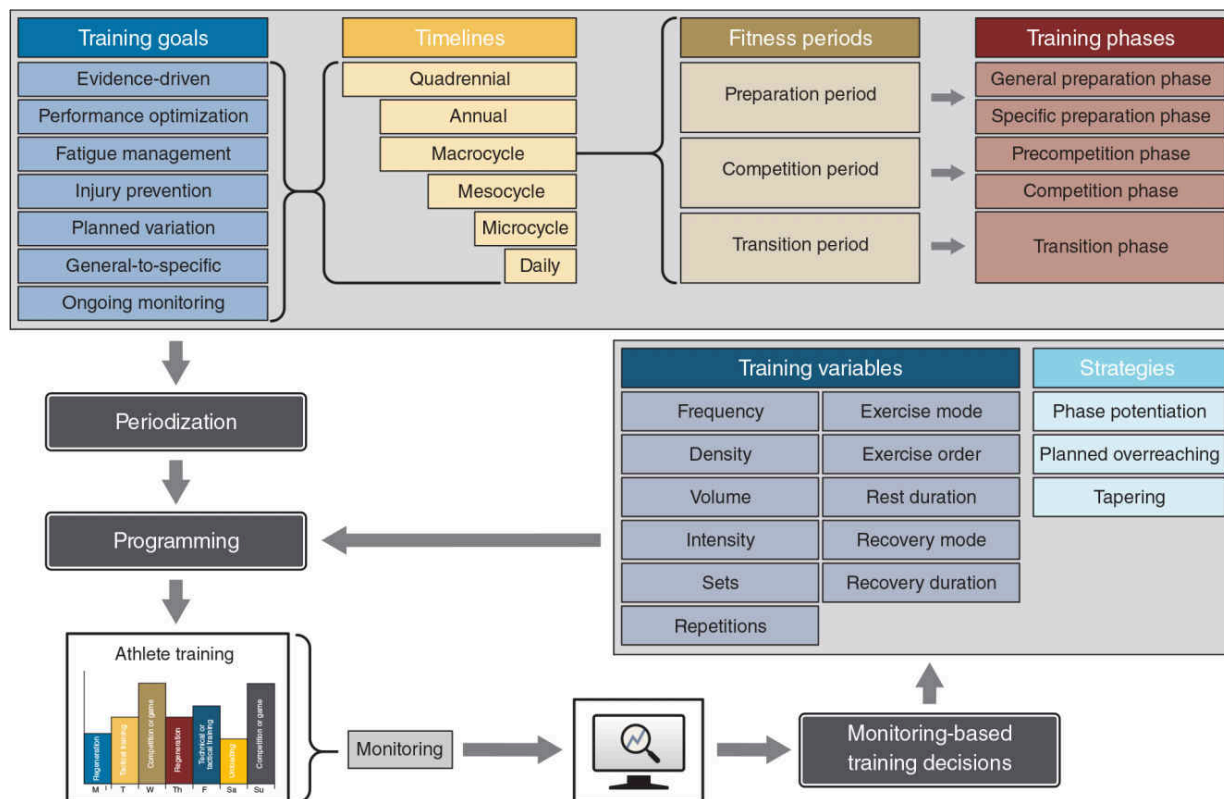


FIGURE 22.1 The symbiotic relationship between periodization, monitoring, and programming.

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This chapter discusses the concept of periodization and its application within a strength and conditioning program. To understand periodization theories and how they are applied to training program design, it is essential to develop an understanding of how the body responds to training (i.e., stressors) (34, 37); this topic is discussed first. Second, the basic models that are used to plan the training process are discussed. Third, the basic hierarchal structure of a periodized training program is discussed to demonstrate how the training year is broken into smaller blocks of training, each with its own training goals and priorities. It is important to note that this overall schedule of training encompasses all aspects of the athletes' training program, including general conditioning, sport-specific activities, and resistance training. Finally, the second half of this chapter presents detailed examples of a yearlong periodized strength and conditioning program. To understand the intricacies of programming, the reader is encouraged to read [chapter 18](#) first.

Central Concepts Related to Periodization

A successful training program allows for management of the adaptive and **recovery** responses to specific interventions that are delivered in a structured way (41). The ultimate success of any training program centers on its ability to induce specific physiological adaptations and translate those adaptations into increases in performance. At the center of this process is the ability to manage the adaptive response, handle accumulated fatigue, and capitalize on the aftereffects established from the various training factors encountered. The strength of a periodized training plan lies in its ability to sequence and structure the training interventions in order to manage all of these factors and peak performance at appropriate time points (35, 37, 38, 41). Ultimately, peak performance can be optimized only for short periods of time (7-14 days), and the average time it can be maintained is inversely related to the average intensity of the training plan (35, 36). In order to elucidate how periodized training models can manage these factors, three basic mechanistic theories have been established: the **general adaptation syndrome (GAS)**, **stimulus–fatigue–recovery–adaptation theory**, and the **fitness–fatigue paradigm** (35, 38, 41).

Periodization is the logical and systematic process of sequencing and integrating training interventions in order to manage the training process and to increase the potential of achieving peak performance at appropriate time points.

The General Adaptation Syndrome

In the 1930s, Hans Selye, a pioneering researcher on the biological effects of exposure to stressful stimuli, presented the basic concepts of the GAS in which a three-stage response to stress (alarm, resistance, and exhaustion) was presented (82). His later work expanded on these early findings by taking into consideration both physical (9, 84, 85) and psychoemotional stress (20, 83, 86). While not originally conceptualized in the context of physical training, over time the GAS has become one of the foundational concepts used to explain periodization theories (20, 29, 35). Any time the body experiences a novel, new, or more intense stress than previously applied (e.g., lifting a heavier training load or a greater volume-load; see [chapter 18](#)), the initial response, or alarm phase, is marked by an accumulation of fatigue, soreness, stiffness, or reduction in energetic stores that results in a reduction in performance capacity (20, 92). Depending on the magnitude of the stress encountered by the athlete, this response may last several hours, days, or weeks. After this initial response, the body moves into the

resistance phase, when it adapts to the stimulus and returns to a normal functional capacity. If the training stress is appropriately structured and not excessive, these adaptive responses can result in specific biochemical, structural, and mechanical adjustments that further elevate the athlete's performance capacity, resulting in what is termed **supercompensation** (91).

An additional factor that modulates the stress response to training is the athlete's psychological and emotional state (95, 96). If the athlete has chronic elevations in psychological and emotional stress, the magnitude of the fatigue response will be increased during the alarm phase and the time course for recovery will be lengthened, resulting in an elongation of the resistance phase, ultimately resulting in a delayed supercompensation response (38). As such, the strength and conditioning professional must not only monitor the training loads the athlete is exposed to but must also consider the athlete's current psychological and emotional state when manipulating training loads within the periodized training plan. In this regard, the integration of a comprehensive monitoring program can provide additional information that allows that strength and conditioning professional to better manage the athlete's exposure to training stress and increase the potential for appropriate elevations in performance.

If, however, the stress persists (either physical or psychoemotional) for an extended period of time, the athlete can move into the exhaustion phase. If this occurs, the athlete is demonstrating an inability to adapt to the imposed stressors and will present some of the same symptoms noted in the alarm phase. Ultimately, when athletes reach the exhaustion phase, they are most likely experiencing overreaching or overtraining responses (see [chapter 24](#) for additional information) (20). From a training perspective, excessive loading, monotonous training, and overly varied training can all result in the occurrence of the exhaustion phase. Additionally, the responses to training can be affected by other nontraining-related stress (e.g., occupational issues, insufficient sleep, relationship, poor diet) that can contribute to the overall stress level experienced by the athlete. Ultimately, the strength and conditioning professional should strive to avoid the occurrence of this phase of the GAS through the proper planning, management (periodization), and monitoring of training stressors. Although the actual dimensions (i.e., slope, magnitude, and timing) of the curve shown in [figure 22.2](#) are highly individualized, this figure represents the basic application of the GAS to training responses.

Stimulus–Fatigue–Recovery–Adaptation Theory

The stimulus–fatigue–recovery–adaptation theory is an extension of the GAS and suggests that training stimuli produce a general response ([figure 22.3](#)) that is influenced by the overall magnitude of the training stressor (38, 92). Specifically, the greater the overall magnitude of the workload encountered, the more fatigue

accumulates and the longer the delay before complete recovery and adaption can occur. As the athlete recovers from and adapts to the training stimuli, fatigue dissipates, and preparedness and performance increase. If no new training stimulus is introduced, a state of involution or detraining (i.e., a reduced overall capacity, to below the current baseline) is observed. In contrast, if a new training stimulus is introduced, the process is repeated. This basic pattern is present whenever an athlete is exposed to a training exercise, session, day, or cycle within a periodized training plan. It should be noted that while recovery is an important part of the training process, it is not always necessary to reach a state of complete recovery before engaging in a new bout or session of training (72). The manipulation of workloads and training intensities through use of light and heavy sessions or days of training can be used to modulate fatigue and recovery responses (8, 26, 88) while allowing for fitness to be either increased or maintained. Conceptually, this theory serves as the foundation for sequential periodization models in that these models allow for the manipulation of various training factors to modulate the athlete's overall fatigue levels, rate of recovery, and adaptive response to the training stimuli.

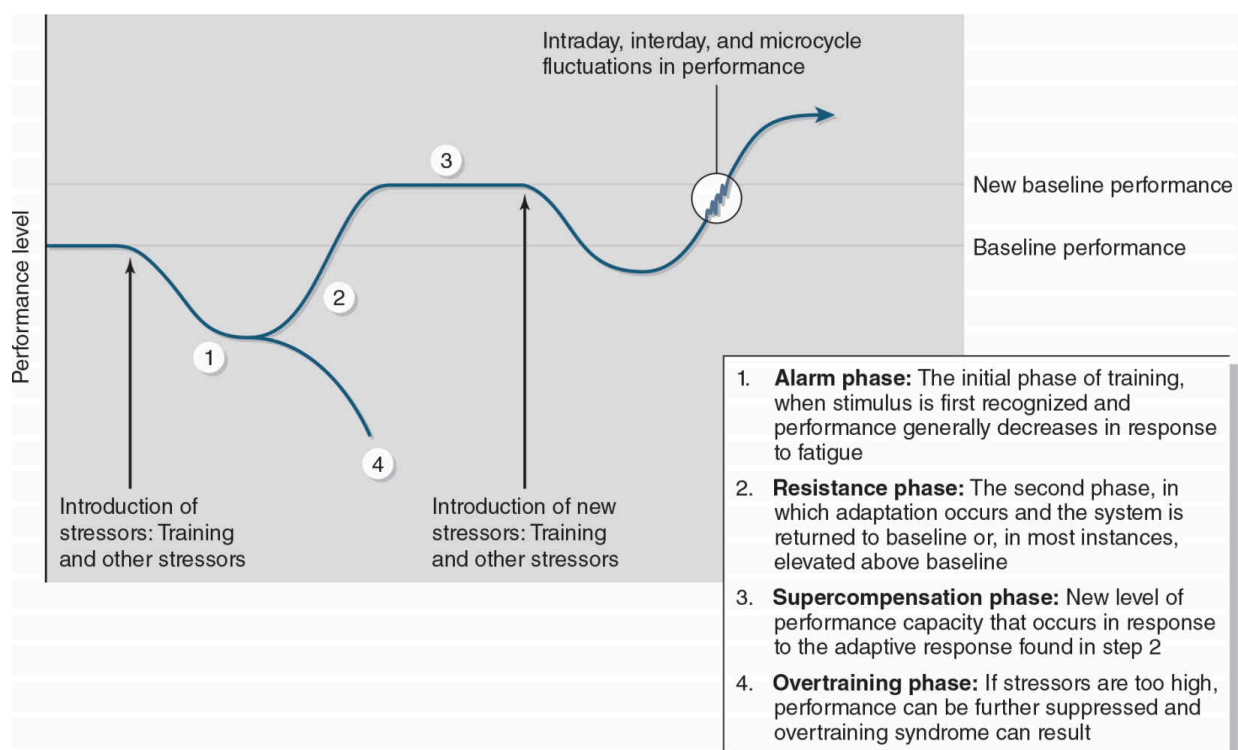


FIGURE 22.2 The general adaptation syndrome (GAS) and application to periodization.

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Fitness–Fatigue Paradigm

Generally, a summation of the two primary training aftereffects (i.e., fitness and fatigue) occurs in response to training interventions that influence the athlete's level of preparedness (15, 38, 107). In 1995, Zatsiorsky (106) presented the classic explanation of these relationships as the two-factor model, more commonly referred to as the *fitness–fatigue paradigm* ([figure 22.4](#)). Ultimately, every training bout, session, or cycle creates both fatigue and fitness aftereffects, which summate to create a state of preparedness (15, 106). When training loads are the highest, fitness becomes elevated, but because of the high training loads, a concomitant increase in fatigue occurs. When fitness and fatigue are summed in this case, the level of fatigue results in a reduction in preparedness. On the other hand, when training workloads are low, little fatigue occurs and minimal fitness is developed, resulting in a low level of preparedness. Thus, the sequencing of training loads becomes important in that it allows for training workloads to be varied in a systematic manner. An important thing to remember is that fatigue dissipates at a rate two-thirds faster than fitness, thus allowing preparedness to become elevated if appropriate training strategies are used to retain fitness while reducing fatigue (35, 38, 41). While the fitness–fatigue paradigm is classically represented as one fatigue, fitness, and preparedness curve, it is likely that each training factor stimulates its own individual fitness, fatigue, and preparedness aftereffect response (15, 38, 41). These aftereffects are often considered to be residual training effects and serve as a fundamental concept underlying the use of sequential periodization models (35, 41). Ultimately, the residual training effects of one training period have the potential to affect the level of preparedness in subsequent training periods, depending on the overall structure of the periodized training plan (35, 41).

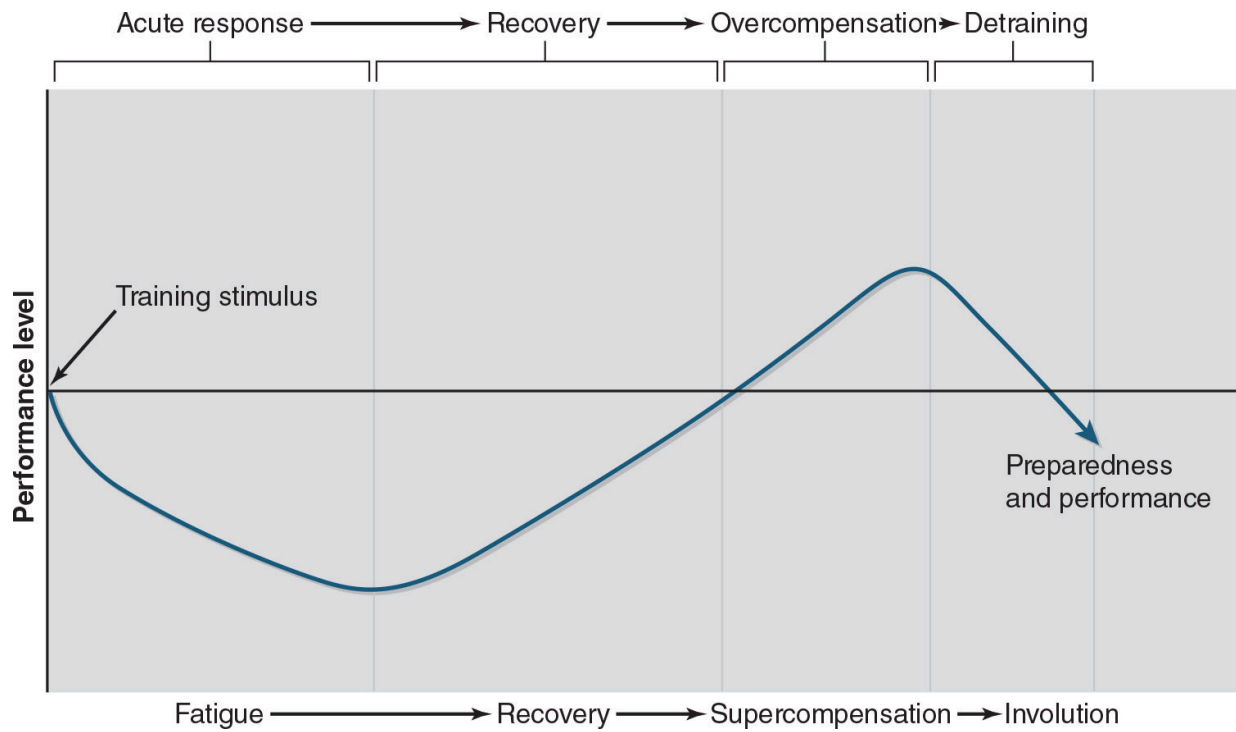


FIGURE 22.3 The stimulus–fatigue–recovery–adaptation theory, with interchangeable terminology.

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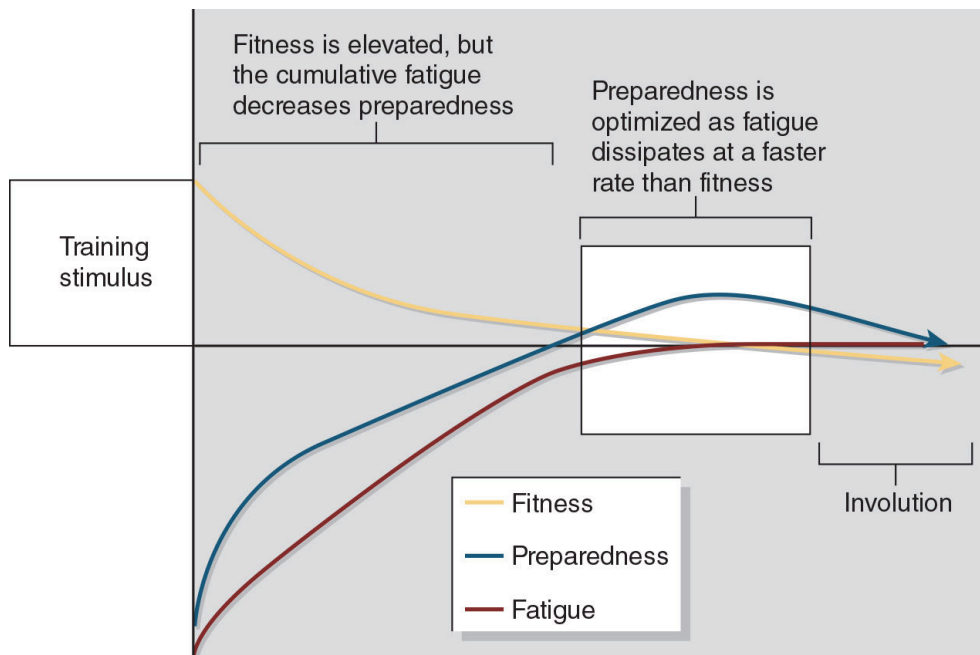


FIGURE 22.4 The fitness–fatigue paradigm.

Adapted by permission from G.G. Haff and E.E. Haff, "Training Integration and Periodization," in *NSCA's Guide to Program Design*, 2nd ed., edited for the National Strength and Conditioning Association by M.T. Jones (Human Kinetics, 2024), 260. Adapted from Stone et al. (2007) and V.M. Zatsiorsky, W.J. Kraemer, and A.C. Fry, *Science and Practice of Strength Training*, 3rd ed. (Human Kinetics, 2021).

Periodization and Planning the Training Process

The development of a periodized training plan is based on three interdependent levels: periodization, planning, and programming ([table 22.1](#)).

TABLE 22.1 Interrelationship of Periodization, Planning, and Programming

Management level	Structure	General description	Specific description
Macro-management	Periodization	The bird's-eye overview of the program direction and goals of the training process; typically depicted as the annual or multiyear training plan	Uses the competitive schedule to create the scaffold from which training is planned and implemented
↓	Planning	Provides the model of periodization used to guide the training process and programmatic decisions	Applies a training model: parallel, sequential, or emphasis model
	Programming	Defines the structure of the actual training interventions (modes and methods)	Structures the training plan and provides the modes and methods used to create the training program
Micro-management			

Adapted from Haff (2017); Haff (2021); Haff (2024).

When beginning the construction of a periodized training plan the first level, or periodization level, provides the scaffold from which the training process is developed (33, 35, 38). At this level, based on the competitive schedule, the annual training plan is subdivided into various periods and phases to provide a basic framework from which the detailed training plan is established. At the second level, or planning level, a training model (i.e., parallel, sequential, or emphasis model) is selected to guide programmatic decisions. Finally, once the models of training are selected and aligned with the various periods and phases contained within the annual plan, they can be used to inform the third level of the process, which involves the actual construction of the training program (41). The programming level is where the various modes, methods, and targeted training strategies are implemented and the actual training loads (i.e., intensities and volumes) and training structures (e.g., loading patterns and set types) are prescribed (35, 41).

Models of Periodization

While the scientific and coaching literature presents numerous models of periodization, all models can be classified under three general models ([figure 22.5](#)).

Parallel Training Models

The **parallel training models**, as shown in [figure 22.5](#), involve targeting all training factors simultaneously within one training session, training day, series of training days, or a microcycle (33, 35, 37, 38). Sometimes this approach is referred to as the *concurrent* or *complex parallel approach* (4). This planning approach aligns with the early stages of most long-term athlete development models (61, 62) where youth athletes are exposed to multilateral training strategies. Additionally, this approach works well with novice and developmental athletes (38). A major concern with parallel models is that as the athlete develops his or her total training load (i.e., volume or load, time dedicated to training, etc.) must be continually increased (35) and may exceed his or her training tolerance (51). As such, this model may not be the best option when working with intermediate to elite athletes, who require greater training variation and stimulus to continue to stimulate adaptation and elevate performance (16, 35, 37, 38). With these athletes it may be warranted to implement different planning strategies (i.e., emphasis or parallel models) to optimize the training process (38).

Sequential Training Models

Sequential training models, as shown in [figure 22.5](#), were created to address the limitations associated with the parallel model of training, such as the inability to simultaneously develop multiple motor and technical abilities, the need for longer preparation periods, and the inability to accommodate dense competitive schedules (38, 52). These models are used to organize an individual or limited number of training factors into a logical pattern that guides the athlete toward a targeted outcome. This approach is ideally suited for athletes who require a greater training stimulus than what can be provided with the parallel approach (38).

This approach to organizing training has been successfully implemented with individual sports such as weightlifting (97), cross-country skiing (87), cycling (23, 80), track and field (74), and aerobic endurance sports (53). Additionally, this approach has been successfully implemented with team sports such as American football (46), basketball (76), football (30), ice hockey (81), and team handball (63).

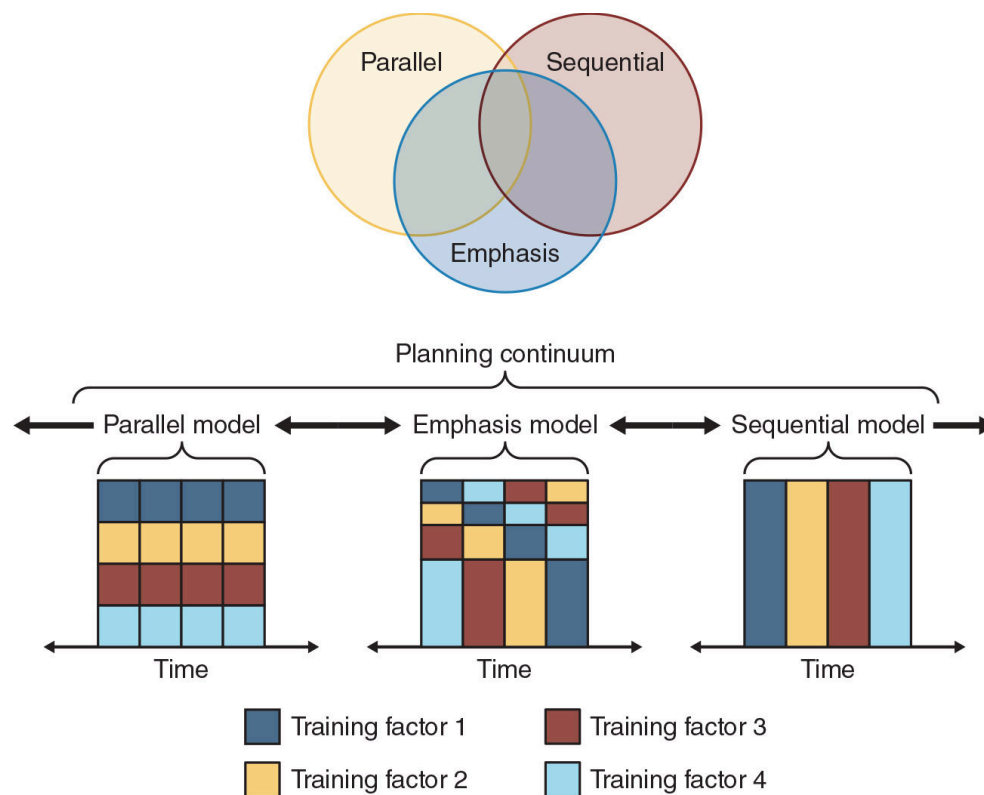


FIGURE 22.5 Three basic models of planning: parallel, emphasis, and sequential.

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Emphasis Training Models

Emphasis training models, as shown in [figure 22.5](#), contain aspects of both parallel and sequential models of periodization (35, 37, 38). Similar to parallel models, multiple training factors are targeted simultaneously but with varying degrees of emphasis over time (38). Over time the degree of focus on the individual training is shifted in a sequential manner that aligns with the periodized training plan. Generally, when this model is implemented the training emphasis is rotated every two weeks to optimize performance capacity (108). When varying the emphasis some training factors will be undertaken with training loads that stimulate adaptations, while some training factors are undertaken with training loads that are used to maintain the athlete's current state (38, 108). Additionally, some training factors will be targeted with a minimal training dose, which would result in a degree of detraining.

This training model is ideally suited for intermediate to advanced athletes who compete in individual or team sports and have congested competitive schedules (38). Examples of the use of this model can be seen in several individual sports, such as track cycling (70), sprinting (27), and mixed martial arts (19). This model has been also recommended for team-sport athletes (32) as well as tactical athletes.

Periodization Hierarchy

Ultimately, periodization is simply a means of organizing the planning of the training process so that it is partitioned into specific time periods ([table 22.2](#)) (36, 37, 41). The multiyear training plan covers the most time but is the least detailed plan within a periodized training structure. For example, it may involve the basic progression of a college soccer player from her freshman to senior year and contain key developmental goals that are targeted within each year of training. This multiyear training structure is then subdivided into more detailed individual **annual training plans** that are developed based on the athlete's progression through the various stages or benchmarks forecasted across the multiyear training plan. In sports that have only one competitive season, such as American football, an annual training plan would be represented as a **macrocycle**. However, in a sport such as college track and field, the annual plan would be divided into two macrocycles because of the indoor and outdoor seasons typical of this college sport. Alternatively, if a high school athlete competes in American football in the fall, basketball in the winter, and track and field in the spring, the annual plan would be subdivided into three macrocycles. Typically, the macrocycle lasts several months up to a year, depending on the sport or the number of sports participated in within the year. Within each macrocycle are **mesocycles**, each lasting several weeks to months, typically two to six weeks, with four weeks being the most common duration. The number of mesocycles within each macrocycle depends on the training targets and the length of the macrocycle within the annual training plan. Each mesocycle is then broken down into individual microcycles that last from several days to weeks; the most common duration is one week (35, 38). Within each **microcycle** are training days that are further subdivided into training sessions.

Periodization of training begins with general global training targets set forth in the multiyear or annual training plan and becomes more specific as the program is developed for the macro-, meso-, and microcycles. For example, annual training plans set the general pathway for a training year, while the other cycles provide the structure that is used to guide the means, methods, and modes used to create the training program that is used to guide the athlete's development toward the primary competitive targets.

TABLE 22.2 Periodization Cycles

Period	Duration	Description
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Period	Duration	Description
Multiyear training plans	2-4 years	This contains interlinked annual training plans; the most common multiyear plan is the quadrennial plan.
Annual training plan	1 year	The overall annual training plan can contain a single or multiple macrocycles. It can be subdivided into preparatory, competitive, and transitional periods of training.
Macrocycle	Several months to a year	It contains preparatory, competitive, and transitional periods of training. If it encompasses one year, some authors refer to this as an <i>annual plan</i> .
Mesocycle	2-6 weeks	This medium-sized training cycle is sometimes called a <i>block of training</i> . It consists of microcycles that are linked together. The most common duration is 4 weeks.
Microcycle	Several days to 2 weeks	This small-sized training cycle is comprised of multiple workouts. It can range from several days to 2 weeks in duration; the most common duration is 1 week (7 days).
Training day	1 day	One training day can contain multiple sessions. It is designed in the context of the microcycle in which it is contained.
Training session	Several hours	It generally consists of several hours of training. If the session has >30 min of rest between bouts of training, it should be considered as having multiple workouts.

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Periodization Periods

The overall variation and structure of the program design variables within each individual meso- and microcycle are constructed based on the periods included in the macrocycle or annual training plan (35, 36, 38). Across these periods of the training plan, the volume and intensity of the training and conditioning program generally receive the greatest attention, but the time spent acquiring and perfecting sport-specific technique must also be considered when one is constructing the overall periodized training plan (77). Ultimately, periodized training plans systematically shift training foci from general nonspecific activities of high volume and low intensity toward activities of lower volume and higher intensities over a period of many weeks or months to help reduce the potential for overtraining while optimizing performance capacities.

The basic sequencing of periodized training programs and how training progresses through the various phases for specified training targets are very similar to the sequencing and evolution of learning academic concepts. For example, in academics we start with simple concepts and skills that progress to more complex concepts; that is, we develop simple skills to provide a sound foundation for learning more complex items. Ultimately, the periods within a periodized training plan serve as the pathway for developing foundational physical, psychological, and performance attributes into more complex sport-specific targets.

In the classic periodization literature, the major divisions of training are classified as the preparatory, competitive, and transition periods (34, 66, 89). Stone, O'Bryant, and Garhammer (89) modified this classic model to include a "first transition" between the preparatory and competitive periods of training. Based on this structure, some periodization models often contain four distinct but interrelated training periods: preparatory, first transition, competitive, and second transition. [Figure 22.6](#) presents the basic periodization model described by Stone, O'Bryant, and Garhammer (89). This model is often applied for novice athletes with a lower training status. Generally, in this application, intensity begins lower and gradually increases, while volume starts higher and slowly decreases as the athlete becomes more conditioned. It is important to note that not all novice athletes are able to tolerate large changes in these variables and that smaller fluctuations may be required (89, 98, 99). It is also important to note that even though these fluctuations are often represented graphically as straight lines, the volume and intensity progressions are in fact nonlinear because of the fluctuations in the loadings that occur at the micro- and mesocycle levels (34, 35, 77). This basic misunderstanding of the classic models of periodization has resulted in these types of models being falsely termed *linear periodization models*

(35, 51, 77, 93). Ultimately, the term *linear periodization* creates a false narrative because one of the major tenets of periodization is the removal of linearity from the training process (38, 51, 72, 77, 93).

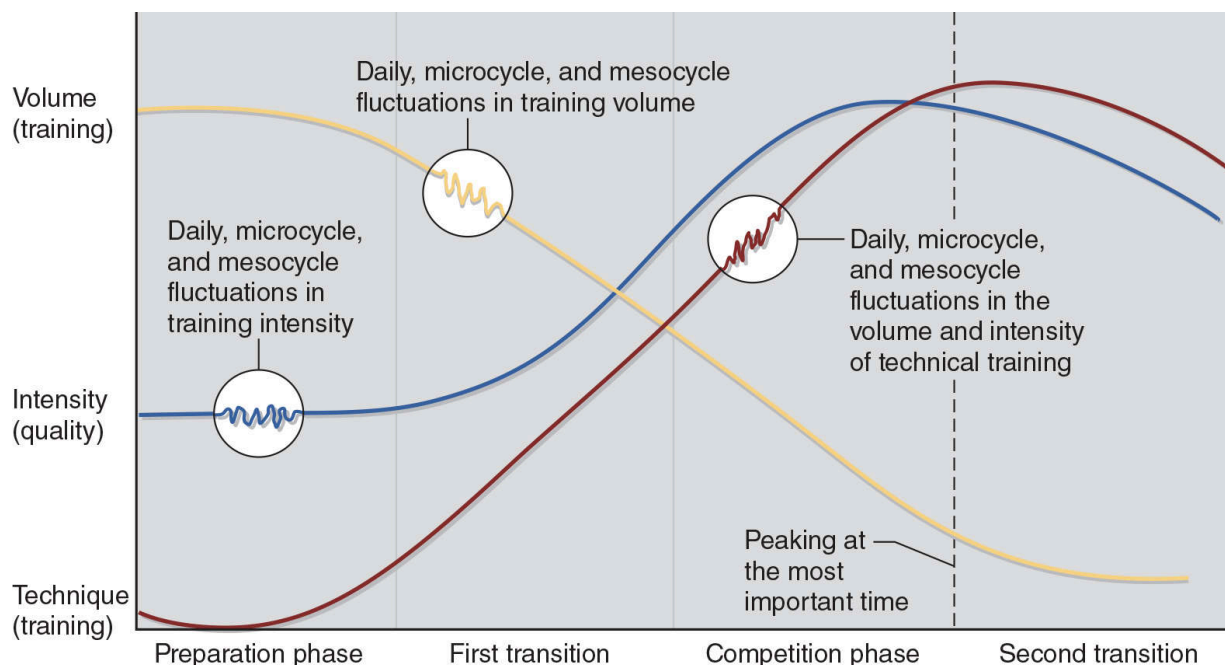


FIGURE 22.6 Matveyev's model of periodization (appropriate for novices).

Adapted by permission from G.G. Haff and E.E. Haff, "Training Integration and Periodization," in *NSCA's Guide to Program Design*, 2nd ed., edited for the National Strength and Conditioning Association by M.T. Jones (Human Kinetics, 2024), 268.; M.H. Stone and H.O. O'Bryant, *Weight Training: A Scientific Approach* (Burgess, 1987).

Advanced athletes tend to train closer to their abilities and have smaller adaptation windows. Therefore, these athletes require greater training variation as well as higher volumes and intensities to allow them to continue to experience appropriate training stimuli (77). For example, Zatsiorsky and colleagues (107) demonstrate that a stimulating load for a novice athlete would be a maintenance load at best for advanced athletes. To address this issue, the shift from higher-volume to higher-intensity training can occur earlier in the preparatory phase, with higher overall training volumes as compared to those in the base model presented in [figure 22.6](#).

Preparatory Period

In outlining a periodized training plan, the starting point is usually the **preparatory period**. This period occurs when there are no competitions, and technical, tactical, or sport-specific work is limited. This period often corresponds to what is termed the *off-season*. The central goal of this period of training is to develop a base level of conditioning to increase the athlete's ability to tolerate

more intense training. Based on the model presented in [figure 22.6](#), the conditioning activities would begin with relatively low intensities and high volumes: long, slow distance running or swimming; low-intensity plyometrics; and high-repetition resistance training with light to moderate resistances. Traditionally, the preparatory period is subdivided into general and specific phases. The **general preparatory phase** typically occurs during the early part of the period and often targets the development of a general physical base (38, 69). This early part of the preparatory period includes high training volumes, low training intensities, and a larger variety of training means that are structured to develop general motor abilities and skills (49, 65). The **specific preparatory phase** occurs after the completion of the general preparatory phase and involves a shift in training focus. From the training base that has been established, this phase expands the athlete's training base through an increased emphasis on sport-specific training activities that prepare the athlete for the competitive period (18, 69). During the preparatory period, resistance training phases can be created to depict more refined differences in training intensity and volume. In order, these are the hypertrophy (or strength–endurance) phase and the basic strength phase (89, 91).

Hypertrophy or Strength–Endurance Phase

The **hypertrophy phase**, which is also referred to as the **strength–endurance phase**, generally occurs during the early portion of the preparatory period (i.e., the general preparatory phase) (24, 40, 41, 93). During this phase, the training intensity is low to moderate, and the overall volume is high. The primary goals during this phase are to enhance work capacity and increase muscle mass. This development serves as the foundation for the higher-intensity training in subsequent phases and periods (42, 43). With strength or power athletes, the primary target might be to stimulate hypertrophic effects while increasing strength–endurance. With aerobic endurance athletes, the primary goal would be to increase strength–endurance without significantly increasing hypertrophy. Regardless of the sport or athlete being trained, it is generally accepted that during the general preparatory phase, sport conditioning activities may not be specific to the athlete's sport. However, as the athlete moves into the specific preparatory phase, over several weeks the training activities will become more sport specific. For example, sprinters may begin the general preparatory phase with longer-distance runs (longer than their competitive distance, not traditional distance running; for example, a 100 m sprinter may do 400 m runs to establish a foundation) at slower speeds in conjunction with lower-intensity plyometrics such as double-leg bounding and hopping, as well as more basic resistance exercises that are not necessarily biomechanically or structurally similar to running (e.g.,

back squat and leg curl). Generally, the athlete performs resistance training with low to moderate intensities for high volumes ([table 22.3](#)).

TABLE 22.3 A Periodized Model for Resistance Training

Period	Preparatory		First transition	Competition		Second transition	
Phase	General preparatory	Specific preparatory	Precompetitive	Main competitive		Postcompetitive	
Season	Off-season		Preseason	In-season		Postseason	
Emphasis	Hypertrophy/strength endurance	Basic strength	Strength/power	Peaking	or	Maintenance	Active rest
Intensity	Low to moderate	High	Low to very high	Very high to very low		Moderate to high	Recreational activities (may not involve resistance training)
	50%-75% 1RM	80%-95% 1RM	87%-95% 1RM ^a	50% to ≥93% 1RM		85%-93% 1RM	
			30%-85% of 1RM ^b				
Volume	High	Moderate to high	Low	Very low	Low to moderate		
	3-6 sets ^c	2-6 sets ^c	2-5 sets ^c	1-3 sets ^c	~2-5 sets ^c		
	8-20 reps	2-6 reps	2-5 reps	1-3 reps	3-6 reps		

^aThese percentages of 1RM apply to nonpower core exercises.

^bThese percentages of 1RM apply to power exercises. The actual percentage used to elicit power development depends on the exercise that is used. For more information, see Kawamori and Haff (55).

^cThese recommendations do not include warm-up sets and represent only target sets for core exercises (40); they also do not include lower-intensity recovery days that are often part of a periodized training plan (40, 89, 92).

Adapted from Haff and Haff (2012); Stone et al. (1981); Stone and O'Bryant (1987); Stone et al. (1982); Stone et al. (2007).

However, it is important to note that throughout this phase, daily variations in training intensity and workload will facilitate recovery (35, 38). Additionally, recovery weeks or microcycles may be placed throughout the phase and most often at the end of the phase before the next phase of training begins.

The hypertrophy or strength–endurance phase involves low to moderate intensity (50%-75% of the 1RM) and high volumes (3 to

6 sets of 8-20 repetitions).

Basic Strength Phase

In the later portion of the preparatory period, during the specific preparatory phase, the primary aim of the **basic strength phase** is to increase the strength of the muscles that are essential to the primary sport movements (12-14). For example, the sprinter's running program would progress to include interval sprints of a moderate distance and more complex and specialized plyometric drills. The resistance training program also becomes more specific to the sport (e.g., squat, power clean, Bulgarian squat) and involves heavier loads performed at lower volumes than in the hypertrophy or strength–endurance phase ([table 22.3](#)). As with the hypertrophy or strength–endurance phase, daily variations in training load facilitate recovery (35, 38).

The basic strength phase involves higher intensity (80%-95% of the 1RM) and moderate to high volumes (2 to 6 sets of 2 to 6 repetitions).

First Transition Period

As originally described by Stone and colleagues (89-91), the **first transition period** is a link between the preparatory and competitive periods. Classically, the resistance training in this period focuses on the development of strength and power as noted in Stone, O'Bryant, and Garhammer's (89) seminal paper on the periodization of strength training. The central aim of this period is to shift the training focus toward the elevation of strength and its translation into power development (89, 90). In order to maximize this process and facilitate recovery, variations are observed in training intensity and workload at the microcycle level (35, 38). Additionally, the last week of the period is marked by reduced volume, intensity, or both to achieve recovery before the beginning of the competitive period.

The main phase within the first transition period is the **strength/power phase**. In this phase, for example, the sprinter's interval and speed training intensifies to near competitive pace; speed drills are performed (e.g., sled towing, sprints against resistance, and uphill sprints); plyometric drills mimic sprinting; and the resistance training program involves performing power and explosive exercises at low to very high loads with low volumes. As explained in [chapter 18](#), the load assignments for power exercises do not follow the typical %RM–repetition relationship, but their relative intensities are elevated during this phase ([table 22.3](#)). Specifically, the exercises selected in this phase can dictate the loading that is used (55). For example, the development of power may be

facilitated with a load of 80% of the 1RM with power cleans performed from the hang (55) or floor (17), while a load of 50% to 70% of the 1RM may be used with the bench press throw when the aim is to maximize power development (55). Ultimately, to address both strength and power development, a mixed training approach is warranted in which heavy- and low-load training is used to optimize both attributes (44).

The strength/power phase involves low to very high loads (30%-95% of the 1RM, depending on the exercise) and low volumes (2 to 5 sets for 2 to 5 repetitions).

Competitive Period

The central training target during the **competitive period** is preparing the athlete for competition by further increasing strength and power via additional increases in training intensity while decreasing volume. This process can be a delicate balancing act, because adequate volume and intensity of training are needed to maintain competitive preparedness, and reductions in volume, intensity, or both are needed to optimize performance. To understand this conundrum, consider the fitness–fatigue paradigm presented earlier. If training workloads (volume or intensity) are reduced too much, fatigue will be reduced, but there will be a concomitant decrease in overall fitness that results in a reduction in competitive preparedness. Also during this period, time spent practicing sport-specific skills and tactics increases dramatically, and a proportional decrease occurs in time spent performing physical conditioning activities such as resistance training. For example, a sprinter places even more emphasis on speed, reaction time, sprint-specific plyometric drills, and technique training. The competitive period may last for one or two weeks for some sports in which a peaking program is employed (6, 36, 69). **Peaking** programs attempt to place the athlete in peak condition for about one or two weeks. Trying to extend this to longer durations ultimately results in reduced performance capacity as a result of reduction in fitness or potential overtraining (36, 38). Depending on the load-reduction strategy employed, peaking programs result in a progressive shift from higher-intensity training toward lower-intensity work designed to reduce fatigue as the athlete moves through the taper before competition (36). As shown in [table 22.3 \(page 730\)](#), resistance training may range between 50% and $\geq 93\%$ of the 1RM, depending on where athletes are in the peaking program.

For team sports, this period spans an entire season and may last for many months, requiring the use of a **maintenance** program (5). Because of the prolonged duration of the competitive period in this situation, the intensity and volume of training must be manipulated on a microcycle basis to maintain

strength and power while managing the fatigue associated with a frequent competition schedule. Generally, a maintenance program is marked by moderate- to high-intensity training (e.g., 85%-93% 1RM) at low to moderate volumes. At the microcycle level, the training loads are modulated based on the training, travel, and competitive schedule. Careful monitoring of the athlete's performance capacity and recovery is critical during the team-sport athlete's competitive period.

The competitive period includes peaking and maintenance. For peaking, athletes use very high to low intensities (50% to $\geq 93\%$ of the 1RM) and very low volume (1 to 3 sets of 1 to 3 repetitions) for one to two weeks. For maintenance, athletes modulate training between moderate and high intensities (85%-93% of the 1RM) with moderate volumes (about 2 to 5 sets of 3 to 6 repetitions).

Second Transition Period (Active Rest)

Between the competitive season and the next annual training plan or the preparatory period of a macrocycle, a **second transition period** is often used to create a linkage (89). This period is sometimes referred to as an **active rest** or **restoration** period and generally lasts for one to four weeks (38). It is important to note that if active rest is extended for a prolonged duration, athletes will require a much longer preparatory period to regain their performance capacities (39). Therefore, it is generally recommended that the second transition period last no longer than four weeks unless an athlete requires additional time to recover from an injury. During this period, aggressive training immediately after the peak performance or end of the maintenance phase should be avoided so athletes can rehabilitate injuries and rest physically and mentally (12-14). For example, a sprinter may engage in recreational activities such as volleyball, racket sports, and swimming in a leisurely manner and perform very low-volume, non-sport-specific resistance training with very low loads. A secondary use of the active rest concept is to structure one-week breaks between long phases (three weeks) or periods of training. The purpose is to create an unloading week to prepare the body for the subsequent phase or period of training. The practice of reducing the training load via the reduction of training intensity and volume is believed by many strength and conditioning professionals to reduce overtraining potential.

The second transition (active rest) provides a period of time in which athletes can rehabilitate injuries and refresh both physically and mentally before beginning a new annual training plan or

macrocycle. This period should not last longer than four weeks, because long periods with reduced training will require the athlete to engage in a longer preparatory period to regain sporting form.

Applying Sport Seasons to the Periodization Periods

In practical terms, periodization involves a logical, systematic variation and integration of training to direct the training responses while managing fatigue and optimizing performance in accordance with the seasonal demands of the sport and athletes. Based on the competitive season, the overall annual plan or macrocycle is structured to sequentially develop specific attributes required by the athlete. To avoid monotony, staleness, and overtraining potential, the training program must involve structured variation of key training variables (e.g., volume, intensity, training frequency, training foci, exercise selection) (41). Classically, most intercollegiate and professional sports have an annual schedule that includes an off-season, preseason, in-season, and postseason. These seasons are easy to relate to the periods in a periodized training model ([figure 22.7](#)).

Off-Season

The off-season should be considered the preparatory period; it typically lasts from the end of the postseason to the beginning of the preseason, which can be about six weeks before the first major competition (although this varies greatly). This preparatory period is subdivided into general and specific preparatory phases that are broken down into mesocycles; these interlink to prepare the athlete for the subsequent competitive season. For example, the athlete may complete several rotations of mesocycles that focus on hypertrophy/strength endurance and basic strength ([figure 22.7](#)). Ultimately, these cyclical rotations are selected based on the sport and the athlete's needs. For example, if an American football player needs to gain muscle mass, more mesocycles that target the hypertrophy phase will be prescribed.

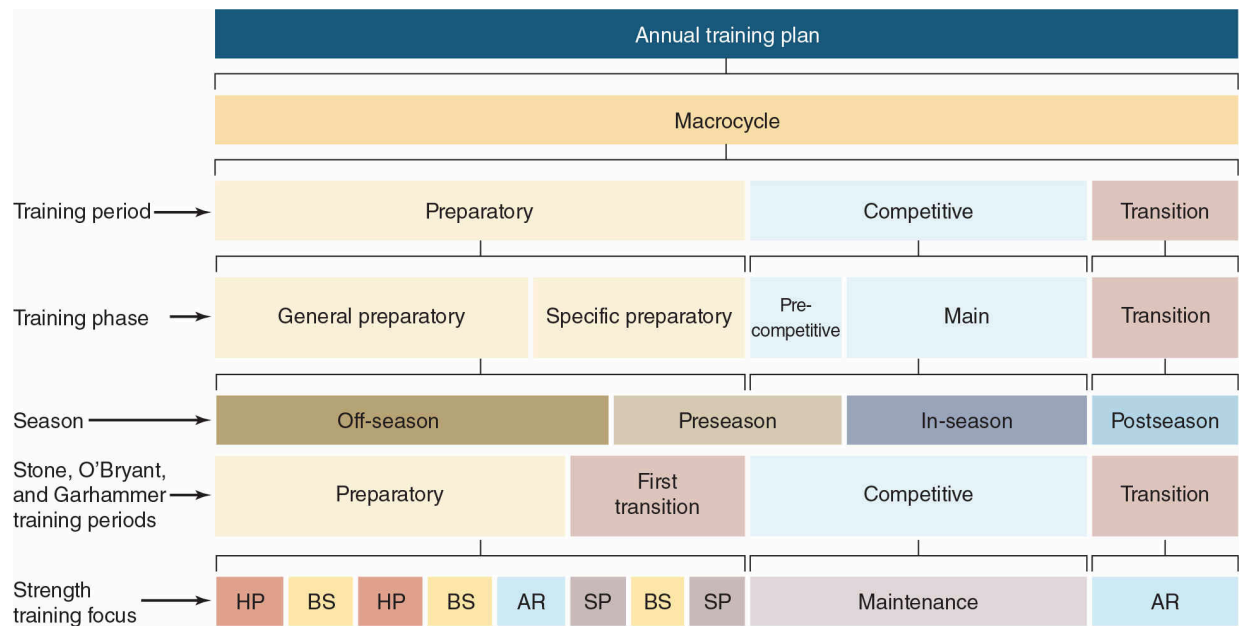


FIGURE 22.7 Relationship of periodization periods to seasons and strength training focus. HP = hypertrophy/strength endurance; BS = basic strength; AR = active rest; SP = strength/power.

Adapted from G.G. Haff, "The Essentials of Periodisation," in *Strength and Conditioning for Sports Performance*, 2nd ed., edited by I. Jeffreys and J. Moody (Routledge, 2021).

Preseason

After the completion of the off-season, the preseason is used to lead into the first major competition. The first transition period is often undertaken at this time with a focus on the strength/power phase of resistance training. This time is used to prepare the athlete for the subsequent competitive period. It is very important to note that the preseason is not the time to build the foundational physical capacities needed for the sport; this should occur primarily in the off-season. The preseason is designed to capitalize on the off-season and elevate the athlete's performance capacity during the competitive period.

In-Season

The competitive, or in-season, period contains all the contests scheduled for the given year, including any tournament games. Most sports have a long season that requires multiple mesocycles arranged around key contests. Thus, a long competitive season (12-16+ weeks) presents unique programming challenges—in these situations, three- or four-week mesocycle blocks that unload the athlete in the last microcycle to allow for fatigue reduction and performance supercompensation before critical contests (41). This does not mean that the athlete is in poor condition for the other contests, because strategically varying where training days occur in conjunction with altering training intensities and volumes across each microcycle can modulate preparedness and elevate performance during congested competitive schedules. Specifically, training

intensity and volume are increased or decreased to maintain physical capacities while reducing fatigue and elevating preparedness before the competitive engagements. During the in-season, monitoring activities are used to make strategic decisions about how to vary training loads to more precisely prescribe training to align with the athlete's current state (11).

Postseason

After the final contest, the postseason or second transition period provides the athlete with a relative or active rest before the beginning of the next year's off-season or preparatory period. It is important to remember that the longer the postseason, the greater the chance for detraining, which results in an increased need for a longer general preparatory phase during the next off-season (38).

Undulating Versus Linear Periodization Models

For better or worse, contemporary periodization literature has adopted the terms *linear* and *nonlinear* when referring to periodization models. However, it should be noted that a central tenet of periodization is the removal of linearity from training (35, 38, 45, 51, 65, 72, 88). In fact, when developing his original model of periodization Matveyev used Yakovlev's concept of supercompensation (104, 105) as the basis for the need for training to be applied in a nonlinear and rhythmic manner (59, 103). Matveyev (65) noted that removing linearity from the training process and implementing appropriate training load variations provides a superior method of training.

Often the traditional resistance training model is falsely referred to as linear due to the gradual and progressive mesocycle increases in intensity over time (7, 51, 77). However, a closer look at the traditional model as described by Stone, O'Bryant, and Garhammer (89, 91) and within the literature (51) shows that the traditional model contains nonlinear variation in training intensity and volume-load at the microcycle level and throughout the mesocycle. Regardless of this, the **nonlinear periodization** model has been introduced in the strength and conditioning industry (2, 58, 79). Alternatively, this model is sometimes described as an undulating or **daily undulating periodization** model because it involves larger daily (i.e., within the microcycle) fluctuations in the load and volume (e.g., repetitions and volume-load) of the assigned core resistance training exercise. For example, when using this programming strategy one might perform 4 sets with a 6RM load (strength focus) on the first day of the week (e.g., Tuesday), 3 sets with a 10RM (hypertrophic focus) on the next training day (e.g., Thursday), and 5 sets with a 3RM load (power focus) on the last training day (e.g., Saturday). In this case the load, volume (e.g., repetitions and volume-load), and training focus are all modified within the microcycle. These modifications are in contrast to what occurs with the **traditional periodization** model, falsely referred to as the **linear periodization** model, according to which the athlete performs the same number of sets and repetitions across the training days and varies the training load. For example, the athlete may perform 4 sets of 6 at 85% of the 1RM on the first day of training, at 75% of the 1RM on the second day, and at 65% of the 1RM on the third day to basically train using a heavy-to-light training structure and introduce variation to the training stimulus. While the traditional model appears to vary only intensity because the repetitions do not change, one must remember that the volume-load changes in this model result in undulations in workload and training volume, supporting the idea that the traditional model is in fact nonlinear and should not be classified as linear (51, 88).

Some research studies suggest that the daily undulating model is more effective than the traditional model (21, 57, 67, 68, 79), although other evidence suggests that no difference exists between these models (2, 10, 31, 78, 100, 101) or that traditional models are superior (47, 48, 74, 75, 101). Proponents of the daily undulating model suggest that one of its strengths is the absence of accumulated neural fatigue caused by extended, ever-increasing training intensities common to the traditional model (56). Conversely, proponents of the traditional model suggest that the overall high volume-loads of training in the undulating model result in greater peripheral fatigue and increased risk of injury because of the high levels of metabolic fatigue this type of programming can stimulate (74, 75). Additionally, examination of the stimulus–fatigue–recovery–adaptation theory ([figure 22.3](#)) and the fatigue–fitness paradigm ([figure 22.4](#)) indicates that the daily undulating model has the potential to decrease athlete preparedness because of accumulated fatigue that occurs with higher volume-load training sessions. Some authors suggest that, based on the GAS, this response may actually increase the potential for overtraining with higher-level athletes (35, 94) and result in an increased injury risk for athletes using this model in conjunction with their sport-based training.

Example of an Annual Training Plan

A sample approach to designing a full annual training plan (i.e., macrocycle for sports with one competitive season) spanning all four sport seasons is presented in [figure 22.8](#).

The annual plan example provided in this chapter defines a macrocycle that lasts from May 1 until April 28. This macrocycle is subdivided into four seasons: off-season (May 1-August 20), preseason (August 21-November 5), in-season (November 6-April 14), and postseason (begins on April 15). Since there is only one competitive season in this annual training plan a single macrocycle structure has been used. This macrocycle is then subdivided into 14 mesocycles, which range in length between two and four weeks. *Note:* The primary focus of this sample annual training plan is to explain how to subdivide the annual plan and to design the resistance training component. Although other modes of training are described briefly (e.g., plyometrics, anaerobic conditioning, aerobic endurance) and can be easily incorporated into the annual plan, this example is not intended to illustrate every aspect of a strength and conditioning program for basketball. For more information on how to structure an annual training plan, consult *NSCA's Guide to Program Design* (41).

Off-Season

Following the postseason active rest period, the athlete should be rested and ready to begin off-season (preparatory period) training. For this example, this preparatory period lasts about 16 weeks, from May 1 until August 20. The goal of this period is to establish a base level of conditioning to increase the athlete's tolerance for more intense training in later phases and periods. In this example, mesocycle 1 is used to expose the athlete to initial strength and conditioning activities at low intensities and volumes to reintroduce them to training. Across the first two microcycles training intensity increases in preparation for performance testing that is planned for microcycle 3. During these first three weeks training loads are applied based on the athlete's rating of perceived exertion or based on of the repetition-in-reserve scale presented in [chapter 18](#). During microcycle 3 the training loads for all non-tested activities will be reduced so that the athletes can partake in a series of initial pretesting to establish the initial training loads for the exercises in the second mesocycle. In later mesocycles, when more or other exercises are added, the training loads can be estimated from loads used in similar exercises or can be determined from RM testing. For example, the strength and conditioning professional can reasonably predict training load for the hang clean from the tested power clean maximum, or the actual RM load could be measured. Other monitoring tests related to

basketball are shoulder and hip flexibility tests, 30-15 intermittent fitness test, 300-yard (274 m) shuttle, line drill, T-test, vertical jump, isometric mid-thigh pull, and skinfold measurements (see [chapter 14](#) for testing protocols).

The resistance training component holds higher priority in the off-season, and an athlete may follow a split routine of four or more training days per week. In this example, the basketball player begins the off-season with a total body, three-times-per-week regimen but soon progresses to four training days per week with variations in training intensity across the microcycle to allow fatigue to be managed. These advancements also involve a gradual increase in loading, with associated decreases in training volume as the athlete shifts his or her training focus toward maximizing strength development. Other training includes sprint and agility training as well as plyometrics and metabolic conditioning. These conditioning workouts are scheduled on nonresistance training days, and flexibility training should be emphasized in the warm-up and cool-down portions of each training session.

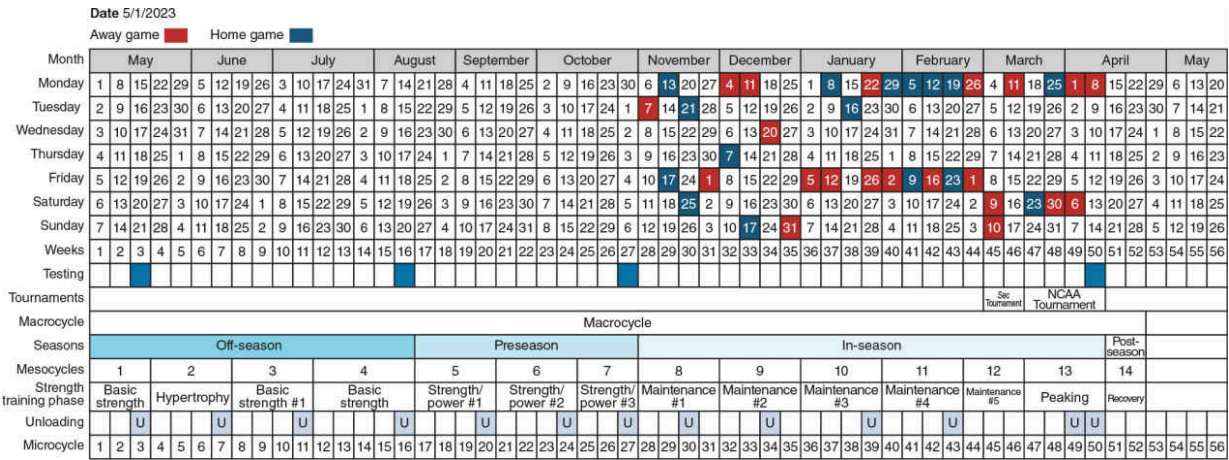


FIGURE 22.8 Sample annual plan for a female basketball athlete.

Preseason

After undertaking one or two unloading weeks following the cessation of the off-season training period, the athlete begins the preseason period of training. For this example, the period covers 11 weeks (August 21-October 5). The goals of the preseason are to increase the intensity of sport-specific training and the attention given to basketball drills and skills. The resistance training portion is planned for three days per week and focuses primarily on strength and power outcomes. Other training modes (e.g., plyometrics and anaerobic conditioning) hold high priority, especially if they directly contribute to basketball training.

In-Season

After an unloading week following the preseason, the athlete is ready to begin the competitive period of training. The in-season period lasts approximately 22 weeks, spanning a time period from November until April (including a four-week tournament period, although it could last longer). The goals for the in-season are to maintain and possibly improve strength, power, flexibility, and anaerobic conditioning. The time constraints of games, skill and strategy practice, and travel result in a lower volume of off-court training activities. Due to multiple games each week, resistance training may be limited to 30 minutes, one to three times per week, consisting of an undulating regimen of varying volume-loads and relative intensities. Power and basketball-specific nonpower core exercises predominate, with assistance exercises added for balance. See the sample in-season program near the end of this chapter. Plyometric sessions should be alternated with resistance training and conducted once or twice weekly, depending on the number of games.

On days when no resistance training is scheduled, 15 to 20 minutes of various short sprint intervals could be performed once or twice weekly during practice. Speed, agility, and other running conditioning can be incorporated into practice time, and flexibility training can be part of the practice and game warm-up and cool-down. Two or three days of rest should be afforded between resistance training, plyometrics, and sprint interval sessions, depending on the game schedule. This period is similar to the previous training period in that the majority of the athlete's time is spent on skill and strategy development, with the remainder devoted to conditioning.

The athlete is in good condition from the previous mesocycles, so she should be able not only to maintain that condition but also to peak again if the team continues in the conference tournament. If the tournament game schedule does not allow for more than one resistance training session per week, the athlete should perform all the week's power and core exercises (if time allows) and omit the assistance exercises. Two examples of specific weekly tournament schedules are presented near the end of this chapter.

Postseason (Active Rest Period)

Following the completion of the competitive season is a (second) transition period of active rest with no formal or structured workouts. For this example, the transition period lasts a month (April 15-May 12). The goal of the period is for the athlete to recuperate physically and psychologically from the long in-season. Recreational games and fitness activities include swimming, jogging, circuit weight training, volleyball, racquetball, and informal basketball. All activities are performed at low intensities with low volumes.

Reviewing the Annual Plan Example

For any periodization model to function optimally, the sport coach and the strength and conditioning professional must plan the program together to share goals and strategies. This is a critical issue in that working together allows all the training factors that the athlete will engage in to be integrated so as to better manage the training stress and modulate fatigue and recovery. Without the cooperation of all involved professionals, optimal performance cannot be fully achieved.

The example in this chapter is a representation of only one periodization model that could be used to structure training for a sample athlete. Other athletes or sports may require subtle to radical variations in the structure presented here. It is important to remember that a multitude of periodization models can be adapted to meet the needs of the various athletes and sports.

Conclusion

Periodization is a process through which the athlete's training is logically and systematically organized to optimize preparedness and increase performance potential for the most important competitions. The annual training plan (year of training) is divided into macrocycles that contain preparatory, competitive, and transition periods. Each period is subdivided into mesocycles that target specific phases of development: hypertrophy/strength endurance, basic strength, or strength/power. Transitions are used to link mesocycles or macrocycles and provide the athlete with an unloading period that enhances recovery. The overall structure of the macrocycle, mesocycle, and individual microcycles is dictated by the individual sport's competitive season and demands.

Off-Season

Female College Basketball Center
16 weeks: May 1-August 20

Mesocycle 1

May 1-May 14

Focus: Basic strength

Microcycles 1-2 (2 weeks)							
Day	Exercise	Week 1			Week 2		
		Sets	Reps	Intensity	Sets	Reps	Intensity
Mon/Fri	Power clean	2	3/1 ^a	5-6 RPE (4-5 RIR)	3	3/1 ^a	7-7.5 RPE (2-3 RIR)
	Back squat	2	5	5-6 RPE (4-5 RIR)	3	5	7-7.5 RPE (2-3 RIR)
	Bench press	2	5	5-6 RPE (4-5 RIR)	3	5	7-7.5 RPE (2-3 RIR)
	Single-leg squat	2	8/8 ^b	5-6 RPE (4-5 RIR)	3	8/8 ^b	7-7.5 RPE (2-3 RIR)
	Lying triceps extension	2	8	5-6 RPE (4-5 RIR)	3	8	7-7.5 RPE (2-3 RIR)
	Glute ham raise	2	10	5-6 RPE	3	10	7-7.5 RPE

^a3/1 represents a cluster set with a 20-second rest between each rep.

^b8/8 = 8 reps per side.

Microcycles 1-2 (2 weeks)							
				(4-5 RIR)			(2-3 RIR)
	Abdominal crunch	2	20		3	20	
Wed	Hang power clean	2	3	5-6 RPE (4-5 RIR)	3	3	7-7.5 RPE (2-3 RIR)
	Clean pull (floor)	2	5	5-6 RPE (4-5 RIR)	3	10	7-7.5 RPE (2-3 RIR)
	Clean grip Romanian deadlift	2	5	5-6 RPE (4-5 RIR)	3	10	7-7.5 RPE (2-3 RIR)
	Lat pulldown	2	5	5-6 RPE (4-5 RIR)	3	10	7-7.5 RPE (2-3 RIR)
	Biceps curl	2	5	5-6 RPE (4-5 RIR)	3	10	7-7.5 RPE (2-3 RIR)
	Abdominal crunch	2	20		3	20	

^a3/1 represents a cluster set with a 20-second rest between each rep.

^b8/8 = 8 reps per side.

Mesocycle 1

May 15-May 21

Focus: Initial pretesting

Microcycle 3 (1 week)	
5RM power exercises	Power clean Power snatch
10RM for assistance exercises	Lat pulldown Biceps curl Lying triceps extension Upright row
5RM for nonpower core exercises	Back squat Bench press
Other testing	Sit and reach Shoulder elevation 30-15 intermittent fitness test Line drill T-test Vertical jump Isometric mid-thigh pull test Body composition (skinfolds)

Mesocycle 2

May 22-June 18
Focus: Hypertrophy

Microcycles 4-6 (3 weeks)						
Session	Exercises	Sets	Reps	Week 4	Week 5	Week 6
Mon/Fri	Back squat	3	10	80%-84% 10RM	85%-89% 10RM	90%-94% 10RM
	Incline bench press	3	10	80%-84% 10RM	85%-89% 10RM	90%-94% 10RM
	Single-leg squat	3	10/10 ^a	80%-84% 10RM	85%-89% 10RM	90%-94% 10RM
	Three-way shoulder	3	10/10/10 ^b	80%-84% 10RM	85%-89% 10RM	90%-94% 10RM
	Glute ham raise	3	10	Add load if possible		
	Abdominal exercise ^c	3	25			
	<i>Note:</i> Friday session should use a load 10% less than Monday's.					
Wed	Hang power clean	3	10	75%-89% 10RM	80%-84% 10RM	85%-89% 10RM
	Clean grip deadlift	3	10	75%-89% 10RM	80%-84% 10RM	85%-89% 10RM
	Clean grip RDL	3	10	75%-89% 10RM	80%-84% 10RM	85%-89% 10RM
	Lat pulldown	3	10	75%-89% 10RM	80%-84% 10RM	85%-89% 10RM
	Abdominal exercise ^c	3	25			
Microcycle 7 (1 week)						
Session	Exercises	Sets	Reps	Week 7		
Mon/Fri	Back squat	3	10	70%-74% 10RM		
	Incline bench press	3	10	70%-74% 10RM		
	Single-leg squat	3	10/10 ^a	70%-74% 10RM		
	Abdominal exercise ^c	3	20			
	<i>Note:</i> Friday session should use a load 10% less than Monday's.					
Wed	Hang power clean	3	10	70%-74% 10RM		

^a10/10 = 10 per leg.

^b10/10/10 = 10 front raise, 10 side lateral raise, 10 reverse fly.

^cAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, crunch, Russian twist.

Microcycles 4-6 (3 weeks)						
	Clean grip deadlift	3	10	70%-74% 10RM		
	Clean grip RDL	3	10	70%-74% 10RM		
	Abdominal exercise ^c	3	20			

^a10/10 = 10 per leg.

^b10/10/10 = 10 front raise, 10 side lateral raise, 10 reverse fly.

^cAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, crunch, Russian twist.

Mesocycle 3

June 19-July 16

Focus: Basic strength

Microcycles 8-10 (3 weeks)						
Session	Exercises	Sets	Reps	Week 8	Week 9	Week 10
Mon/Fri	Power clean	3	5	80%-84% 5RM	85%-89% 5RM	90%-94% 5RM
	Back squat	3	5	80%-84% 5RM	85%-89% 5RM	90%-94% 5RM
	Bench press	3	5	80%-84% 5RM	85%-89% 5RM	90%-94% 5RM
	Lying triceps extension	2	10	75%-79% 10RM	80%-84% 10RM	85%-89% 10RM
	Upright row	2	10	75%-79% 10RM	80%-84% 10RM	85%-89% 10RM
	Abdominal exercise*	3	20			
	<i>Note: Friday session should use a load 10% less than Monday's.</i>					
Wed	Snatch grip deadlift	3	5	75%-79% 5RM	80%-84% 5RM	85%-89% 5RM
	Snatch grip mid-thigh pull	3	5	75%-79% 5RM	80%-84% 5RM	85%-89% 5RM
	Lat pulldown	2	10	70%-74% 10RM	75%-79% 10RM	80%-84% 10RM
	Hammer curl	2	10	70%-74% 10RM	75%-79% 10RM	80%-84% 10RM
	Glute ham raise	3	10	Add weight if possible		
	Abdominal exercise*	3	20			

Microcycle 11 (1 week)						
Session	Exercises	Sets	Reps	Week 11		
Mon/Fri	Power clean	3	5	70%-74% 5RM		
	Back squat	3	5	70%-74% 5RM		
	Bench press	3	5	70%-74% 5RM		

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, crunch, Russian twist.

Microcycles 8-10 (3 weeks)						
	Abdominal exercise*	3	20			
	<i>Note: Friday session should use a load 10% less than Monday's.</i>					
Wed	Snatch grip deadlift	3	5	70%-74% 5RM		
	Snatch grip mid-thigh pull	3	5	70%-74% 5RM		
	Lat pulldown	2	10	60%-69% 5RM		
	Abdominal exercise*	3	20			

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, crunch, Russian twist.

Mesocycle 4

July 17-August 13

Focus: Basic strength

Microcycles 12-13 (2 weeks)						
Session	Exercises	Sets	Reps	Week 12	Week 13	
Mon/Fri	Hang power snatch	5	5	75%-79% 5RM	80%-84% 5RM	
	Back squat	5	5	75%-79% 5RM	80%-84% 5RM	
	Strict press	5	5	75%-79% 5RM	80%-84% 5RM	
	Leg (knee) curl	3	12	70%-74% 12RM	75%-79% 12RM	
	Seated calf raise	3	12	70%-74% 12RM	75%-79% 12RM	
	Abdominal exercise*	3	20			
	<i>Note: Friday session should use a load 10% less than Monday's.</i>					
Wed	Hang power clean	5	5	70%-74% 5RM	75%-79% 5RM	
	Clean pull (floor)	5	5	70%-74% 5RM	75%-79% 5RM	
	Bent-over row	5	5	70%-74% 5RM	75%-79% 5RM	
	Barbell biceps curl	3	12	65%-69% 12RM	70%-74% 12 RM	
	Triceps pushdown	3	12	65%-69% 12RM	70%-74% 12 RM	
	Abdominal exercise*	3	25			
Microcycles 14-15 (2 weeks)						
Session	Exercises	Sets	Reps	Week 14	Week 15	
Mon/Fri	Hang power snatch	3	5	85%-89% 5RM	90%-94% 5RM	
	Back squat	3	5	85%-89% 5RM	90%-94% 5RM	

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, crunch, Russian twist.

Microcycles 12-13 (2 weeks)						
	Strict press	3	5	85%-89% 5RM	90%-94% 5RM	
	Leg (knee) curl	3	8	75%-79% 8RM	80%-84% 8RM	
	Seated calf raise	3	8	75%-79% 8RM	80%-84% 8RM	
	Abdominal exercise*	3	20			
	<i>Note:</i> Friday session should use a load 10% less than Monday's.					
Wed	Hang power clean	3	5	80%-84% 5RM	85%-89% 5RM	
	Clean pull (floor)	3	5	80%-84% 5RM	85%-89% 5RM	
	Bent-over row	3	5	80%-84% 5RM	85%-89% 5RM	
	Barbell biceps curl	3	8	70%-74% 8RM	75%-79% 8RM	
	Triceps pushdown	3	8	70%-74% 8RM	75%-79% 8RM	
	Abdominal exercise*	3	20			

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, crunch, Russian twist.

Mesocycle 4

August 14-August 20
Focus: Initial pretesting

Microcycle 16 (1 week)	
3RM power exercises	Power clean Power snatch
8RM for assistance exercises	Lat pulldown Biceps curl Lying triceps extension Upright row
3RM for nonpower core exercises	Back squat Bench press
Other testing	Sit and reach Shoulder elevation 30-15 intermittent fitness test Line drill T-test Vertical jump Isometric mid-thigh pull test Body composition (skinfolds)

Preseason

Female College Basketball Center
11 weeks: August 21-November 5

Mesocycle 5

August 21-September 17
Focus: Strength/power 1

Microcycle 17 (1 week)						
Session	Exercises	Sets	Reps	Week 17		
Mon/Fri	Push jerk	5	5	75%-79% 5RM		
	Back squat	5	5	80%-84% 5RM		
	Incline bench press	5	5	80%-84% 5RM		
	Biceps curl	3	10	75%-79% 10RM		
	Dip	3	10	Add weight if possible		
	Abdominal exercise ^a	3	20			
	<i>Note: Friday session should use a load 15% less than Monday's.</i>					
Wed	Power clean	5	5/1 ^b	80%-84% 1RM		
	Clean pull (floor)	5	5	80%-84% 5RM		
	Romanian deadlift	5	5	75%-79% 5RM		
	Pendlay row	3	10	75%-79% 10RM		
	Pull-up	3	10	Add weight if possible (use band if needed)		
	Abdominal exercise ^a	3	20			
Microcycles 18-19 (2 weeks)						
Session	Exercises	Sets	Reps	Week 18	Week 19	
Mon	Push jerk	3	5	80%-84% 5RM	85%-89% 5RM	
	Back squat	3	5	85%-89% 5RM	90%-94% 5RM	
	Incline bench press	3	5	85%-89% 5RM	90%-94% 5RM	
	Biceps curl	3	8	80%-84% 8RM		
	Dip	3	10	Add weight if possible		
	Abdominal exercise ^a	3	20			
Wed	Power clean	3	5/1 ^b	80%-84% 1RM	85%-89% 1RM	
	Clean pull (floor)	3	5	85%-89% 5RM	90%-94% 5RM	

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bCluster set with 20 s between reps.

Microcycle 17 (1 week)						
	Romanian deadlift	3	8	85%-89% 5RM	90%-94% 5RM	
	Pendlay row	3	10	80%-84% 10RM	85%-89% 10RM	
	Pull-up	3	10	Add weight if possible (use band if needed)		
	Abdominal exercise ^a	3	20			
Fri	Push jerk	3	5	75%-79% 5RM	80%-84% 5RM	
	Quarter back squat (from pins)	3	5	80%-84% 5RM	85%-89% 5RM	
	Closed-grip bench press	3	5	75%-79% 5RM	80%-84% 5RM	
	Hammer curl	3	8	75%-79% 8RM	80%-84% 8RM	
	Triceps pushdown	3	8	75%-79% 8RM	80%-84% 8RM	
	Abdominal exercise ^a	3	20			
Microcycle 20 (1 week)						
Session	Exercises	Sets	Reps	Week 20		
Mon/Fri	Push jerk	3	3	75%-79% 3RM		
	Back squat	3	3	75%-79% 3RM		
	Incline bench press	3	3	75%-79% 3RM		
	Biceps curl	3	8	70%-74% 8RM		
	Lying triceps extension	3	8	70%-74% 8RM		
	Abdominal exercise ^a	3	20			
	<i>Note: Friday session should use a load 10% less than Monday's.</i>					
Wed	Power clean	3	3/1 ^b	75%-79% 1RM		
	Clean pull (floor)	3	3	75%-79% 3RM		
	Romanian deadlift	3	3	75%-79% 3RM		
	Pendlay row	3	5	75%-79% 5RM		
	Pull-up	3	8	Add weight if possible (use band if needed)		
	Abdominal exercise ^a	3	20			

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bCluster set with 20 s between reps.

Mesocycle 6

September 18-October 15

Focus: Strength/power 2

Microcycle 21 (1 week)						
Session	Exercises	Sets	Reps	Week 21		
Mon	Hang power clean	5	3	80%-84% 3RM		
	Front squat	3	5	85%-89% 5RM		
	Clean pull (thigh block)	3	5	80%-84% 5RM		
	Push press	3	5	75%-79% 5RM		
	Dip	3	8	Add weight if possible		
	Abdominal exercise ^a	3	20			
Wed	Overhead squat	5	3	75%-79% 3RM		
	Hang power snatch	5	3	75%-79% 3RM		
	Snatch pull (floor)	3	5	85%-89% 5RM		
	Snatch grip Romanian deadlift	3	5	85%-89% 5RM		
	Pull-up	3	10	Add weight if possible (use band if needed)		
	Abdominal exercise ^a	3	20			
Fri	Push jerk	5	3	70%-74% 3RM		
	Jump squat ^b	5	3	45%-49% 1RM		
	Quarter front squat	3	5	80%-84% 5RM		
	Closed-grip bench press	3	5	75%-79% 5RM		
	Glute ham raise	3	10	Add weight if possible		
	Abdominal exercise ^a	3	20			
Microcycles 22-23 (2 weeks)						
Session	Exercises	Sets	Reps	Week 22	Week 23	
Mon	Push jerk	4	3	85%-89% 3RM	90%-94% 3RM	
	Hang power clean	4	3	80%-84% 3RM	85%-89% 3RM	
	Front squat	5	3	90%-94% 3RM	80%-84% 3RM	
	Clean pull (floor)	5	3	80%-84% 3RM	85%-89% 3RM	
	Dip	3	8	Add weight if possible		
	Abdominal exercise ^a	3	20			
Wed	Snatch grip jump shrug ^c	4	3	45%-49% 1RM	50%-54% 1RM	
	Hang power snatch	4	3	70%-74% 3RM	75%-79% 3RM	

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bLoad based off back squat 1RM.

^cLoad based off power snatch 1RM.

Microcycle 21 (1 week)						
	Snatch pull (thigh block)	5	3	85%-89% 3RM	90%-94% 3RM	
	Snatch grip Romanian deadlift	5	3	80%-84% 3RM	85%-89% 3RM	
	Pull-up	3	10	Add weight if possible (use band if needed)		
	Abdominal exercise ^a	3	20			
Fri	Push jerk	4	3	70%-74% 3RM	75%-79% 3RM	
	Jump squat ^b	4	3	45%-49% 1RM	50%-54% 1RM	
	Quarter front squat	5	3	80%-84% 5RM	85%-89% 5RM	
	Closed-grip bench press	5	3	75%-79% 5RM	80%-84% 5RM	
	Glute ham raise	3	10	Add weight if possible		
	Abdominal exercise ^a	3	20			
Microcycle 24 (1 week)						
Session	Exercises	Sets	Reps	Week 24		
Mon	Hang power clean	3	2	70%-74% 2RM		
	Front squat	3	3	75%-79% 3RM		
	Clean pull (thigh block)	3	3	80%-84% 3RM		
	Push press	3	3	75%-79% 3RM		
	Dip	3	8	Add weight if possible		
	Abdominal exercise ^a	3	20			
Wed	Snatch grip jump shrug ^c	3	2	50%-54% 1RM		
	Hang power snatch	3	2	70%-74% 2RM		
	Snatch pull (thigh block)	3	3	80%-84% 3RM		
	Snatch grip Romanian deadlift	3	3	75%-79% 3RM		
	Pull-up	3	8	Add weight if possible (use band if needed)		
	Abdominal exercise ^a	3	20			
Fri	Push jerk	3	2	70%-74% 2RM		
	Jump squat ^b	3	2	50%-54% 1RM		
	Quarter front squat	3	3	80%-84% 3RM		
	Closed-grip bench press	3	3	80%-84% 3RM		
	Glute ham raise	3	8	Add weight if possible		

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bLoad based off back squat 1RM.

^cLoad based off power snatch 1RM.

Microcycle 21 (1 week)

	Abdominal exercise ^a	3	20			
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^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bLoad based off back squat 1RM.

^cLoad based off power snatch 1RM.

Mesocycle 7

October 16-October 29
Focus: Strength/power 3

Microcycle 25 (1 week)

Session	Exercises	Sets	Reps	Week 25		
Mon/Fri	Hang power clean	3	5	70%-74% 5RM		
	Back squat + box jump	3	5+5	80%-84% 5RM		
	Bench press + medicine ball chest pass	3	5+5	80%-84% 5RM		
	Push jerk	3	5	70%-75% 5RM		
	Lying triceps extension	3	8	75%-79% 8RM		
	Abdominal exercise*	3	20			
	<i>Note:</i> Friday session should use a load 10% less than Monday's.					
Wed	Hang power snatch	3	5	70%-75% 5RM		
	Snatch pull (floor)	3	5	80%-84% 5RM		
	Lat pulldown	3	8	75%-79% 8RM		
	Bent-over row	3	8	75%-79% 8RM		
	Glute ham raise	3	10	Add weight if possible		
	Abdominal exercise*	3	20			

Microcycle 26 (1 week)

Session	Exercises	Sets	Reps	Week 26		
Mon/Fri	Hang power clean	4	3	75%-79% 3RM		
	Back squat + box jump	4	3+3	85%-89% 3RM		
	Bench press + medicine ball chest pass	4	3+3	80%-84% 3RM		
	Push jerk	4	3	80%-84% 3RM		
	Lying triceps extension	3	8	80%-84% 3RM		
	Abdominal exercise*	3	20			

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Microcycle 25 (1 week)						
	<i>Note:</i> Friday session should use a load 10% less than Monday's.					
Wed	Hang power snatch	4	3	75%-79% 3RM		
	Snatch pull (floor)	4	3	85%-89% 3RM		
	Lat pulldown	3	8	80%-84% 8RM		
	Bent-over row	3	8	80%-84% 8RM		
	Glute ham raise	3	10	Add weight if possible		
	Abdominal exercise*	3	20			

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Mesocycle 7

October 30-November 5

Focus: Post preseason testing (before in-season)

Microcycle 27 (1 Week)	
2RM power exercise	Power clean Hang power snatch Push jerk Push press
2RM for nonpower core exercises	Back squat Front squat Bench press
Other testing*	Sit and reach Shoulder elevation 30-15 intermittent fitness test Line drill T-test Vertical jump Isometric mid-thigh pull test Body composition (skinfolds)

*Other exercises to test based on in-season program.

In-Season

Female College Basketball Center

23 weeks: November 6-April 14 (including tournament play)

Mesocycle 8

November 6-December 3
Focus: Strength maintenance

Microcycles 28-31 (4 Weeks)							
Session	Exercises	Sets	Reps	Training loads			
				Week 28	Week 29	Week 30	Week 31
1	Clean pull (mid-thigh block)	1	4	75%-79% 1RM	80%-84% 1RM	60%-64% 1RM	85%-89% 1RM
	Back squat	1	6	75%-79% 6RM	80%-84% 6RM	65%-69% 6RM	90%-94% 6RM
	Incline bench press	1	6	75%-79% 6RM	80%-84% 6RM	65%-69% 6RM	90%-94% 6RM
	Lateral shoulder raise	1	10	70%-74% 10RM	75%-79% 10RM	70%-74% 10RM	80%-84% 10RM
	Abdominal exercise*	Max in 60 s					
2	Push press	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Quarter front squat (from pins)	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Closed-grip bench press	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Seated low pulley row	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Abdominal exercise*	Max in 60 s					

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Mesocycle 9

December 4-December 31
Focus: Strength and power development

Microcycles 32-35 (4 Weeks)							
Session	Exercises	Sets	Reps	Training loads			
				Week 32	Week 33	Week 34	Week 35
1	Hang power clean	1	3	65%-69% 1RM	70%-74% 1RM	60%-64% 1RM	80%-84% 1RM

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Microcycles 32-35 (4 Weeks)							
	Front squat	1	5	70%-74% 5RM	80% 5RM	60%-64% 5RM	80%-84% 5RM
	Standing shoulder press	1	5	70%-74% 5RM	80% 5RM	60%-64% 5RM	80%-84% 5RM
	Lying triceps extension	1	10	70%-74% 10RM	80% 10RM	65%-69% 10RM	80%-84% 10RM
	Abdominal exercise*	Max in 60 s					
2	Push jerk	1	3	60%-64% 1RM	65%-69% 1RM	60% 1RM	70%-74% 1RM
	Step-up	1	5	65%-69% 5RM	70%-74% 5RM	60% 5RM	75%-79% 5RM
	Closed-grip bench press	1	10	60%-64% 10RM	65%-69% 10RM	60% 10RM	70%-74% 10RM
	Lat pulldown	1	10	60%-64% 10RM	65%-69% 10RM	60% 10RM	70%-74% 10RM
	Abdominal exercise*	Max in 60 s					

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Mesocycle 10

January 1-January 28

Focus: Strength and power development

Microcycles 36-39 (4 Weeks)							
Session	Exercises	Sets	Reps	Training loads			
				Week 36	Week 37	Week 38	Week 39
1	Power clean	1	2	80%-84% 1RM	85%-89% 1RM	90%-94% 1RM	70%-74% 1RM
	Back squat	1	4	80%-84% 4RM	85%-89% 4RM	90%-94% 4RM	65%-69% 4RM
	Bench press	1	4	80%-84% 4RM	85%-89% 4RM	90%-94% 4RM	65%-69% 4RM
	Barbell biceps curl	1	8	80%-84% 8RM	85%-89% 8RM	90%-94% 8RM	65%-69% 8RM
	Abdominal exercise*	Max in 60 s					

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Microcycles 36-39 (4 Weeks)							
2	Hang power snatch	1	3	70%-74% 1RM	75%-78% 1RM	80%-84% 1RM	60%-64% 1RM
	Forward step lunge	1	4	70%-74% 4RM	75%-78% 4RM	80%-84% 4RM	60%-64% 4RM
	Closed-grip bench press	1	8	70%-74% 8RM	75%-78% 8RM	80%-84% 8RM	60%-64% 8RM
	Upright row	1	8	70%-74% 8RM	75%-78% 8RM	80%-84% 8RM	60%-64% 8RM
	Abdominal exercise*	Max in 60 s					

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Mesocycle 11

January 29-February 15

Focus: Strength and power development

Microcycles 40-43 (4 Weeks)							
Session	Exercises	Sets	Reps	Training loads			
				Week 40	Week 41	Week 42	Week 43
1	Snatch grip mid-thigh pull (block)	1	6	75%-79% 1RM	80%-84% 1RM	60%-64% 1RM	85%-89% 1RM
	Back squat	1	6	80%-84% 6RM	85%-89% 6RM	65%-69% 6RM	90%-94% 6RM
	Bench press	1	6	80%-84% 6RM	85%-89% 6RM	65%-69% 6RM	90%-94% 6RM
	Lateral shoulder raise	1	10	75%-79% 10RM	80%-84% 10RM	70%-74% 10RM	85%-89% 10RM
	Abdominal exercise*	Max in 60 s					
2	Push press	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Quarter back squat (from pins)	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Closed-grip bench press	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Seated low pulley row	1	6	70%-74% 6RM	75%-79% 6RM	60%-64% 6RM	80%-84% 6RM
	Abdominal exercise*	Max in 60 s					

*Abdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

Mesocycle 12

February 16-March 17

Focus: Power and strength optimization

Microcycles 44-46 (3 Weeks—SEC Tournament)							
Session	Exercises	Sets	Reps	Training loads			
				Week 44	Week 45	Week 46	
1	Hang power clean	1	3	70%-74% 1RM	75%-79% 1RM	80%-84% 1RM	
	Front squat	1	5	75%-79% 5RM	80%-84% 5RM	85%-89% 5RM	
	Push press	1	5	75%-79% 5RM	80%-84% 5RM	85%-89% 5RM	
	Lying triceps extension	1	10	75%-79% 10RM	80%-84% 10RM	85%-89% 10RM	
	Abdominal exercise ^a	Max in 60 s					
2	Push jerk	1	3	65%-69% 1RM	70%-74% 1RM	75%-79% 1RM	
	Jump squat ^b	1	5	70%-74% 5RM	75%-79% 5RM	80%-84% 5RM	
	Closed-grip bench press	1	10	65%-69% 10RM	70%-74% 10RM	75%-79% 10RM	
	Lat pulldown	1	10	65%-69% 10RM	70%-74% 10RM	75%-79% 10RM	
	Abdominal exercise ^a	Max in 60 s					

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bIntensity based off of 1RM squat.

Mesocycle 13

March 18-April 7

Focus: Power and peaking performance

Microcycles 47-49 (3 Weeks—national tournament)							
Session	Exercises	Sets	Reps	Training loads			
				Week 47	Week 48	Week 49	
1	Jump squat ^b	1	3	40%-44% 1RM	35%-39% 1RM	30%-34% 1RM	
	Quarter back squat	1	3	80%-84% 3RM	75%-79% 3RM	70%-74% 3RM	
	Push jerk	1	3	80%-84% 3RM	75%-79% 3RM	70%-74% 3RM	
	Lying triceps extension	1	8	80%-84% 8RM	75%-79% 8RM	70%-74% 8RM	

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bIntensity based off 1RM bench press.

Microcycles 47-49 (3 Weeks—national tournament)

	Abdominal exercise ^a	Max in 60 s					
2	Jump shrug	1	3	70%-74% 3RM	65%-69% 3RM	60%-64% 1RM	
	Clean pull (thigh block)	1	3	80%-84% 3RM	75%-79% 5RM	70%-74% 3RM	
	Bench throw ^b	1	4	40%-44% 1RM	35%-39% 1RM	30%-34% 1RM	
	Lat pulldown	1	10	70%-74% 8RM	65%-69% 8RM	60%-64% 8RM	
	Abdominal exercise ^a	Max in 60 s					

^aAbdominal exercise of choice: sit-up, GHD sit-up, toes to bar, reverse crunch, or crunch.

^bIntensity based off 1RM bench press.

Mesocycle 13

April 8-April 14

Focus: Postseason testing (after in-season)

Microcycle 50 (1 Week)

2RM power exercise	Power clean Hang power snatch Push jerk Push press
2RM for nonpower core exercises	Back squat Front squat Bench press
Other testing*	Sit and reach Shoulder elevation 30-15 Intermittent Fitness Test Line drill T-test Vertical jump Isometric mid-thigh pull test Body composition (skinfolds)

*Other exercises to test based on in-season program.

Tournament Option A (Two Games)

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Practice (or rest)	Practice	Practice	Practice	Game	Practice	Game
	Resistance training		Resistance training			

Tournament Option B (Two Games)

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Practice (or rest)	Game	Practice	Practice	Practice	Practice	Game
		Resistance training	Plyometric training	Resistance training		

Tournament Option C (Three Games Back-to-Back)

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Practice (or rest)	Practice	Practice	Practice	Practice	Game	Game
	Resistance training		Plyometric training			
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Game	Rest	Practice	Practice	Practice	Practice	Rest
		Resistance training		Resistance training	Plyometric training	

Comments

- Each mesocycle is considered a block of training and contains two or three individual microcycles; unloading mesocycles are strategically placed to occur before the next mesocycle during the off-season and preseason and are at key microcycles during the in-season.
- This program incorporates variations in volume-load, training intensity, and targeted training exercises.
- Repetition maximum loads are not used, as training to failure has been shown to reduce power output and increase the risk of overtraining. Therefore, percentage ranges based off of RM loads are used to deal with the accumulated fatigue associated with in-season training and competitions.
- It is important to note that these are the target sets and that 3 to 5 warm-up sets should be performed to provide an appropriate specific warm-up for the targeted loadings.
- Refer to [table 18.8 \(page 544\)](#) for the relationship between the %RM and the number of repetitions allowed and [table 18.9 \(page 547\)](#) for the estimation of a 1RM.
- Refer to [table 18.15 \(page 561\)](#) for rest period length assignment based on repetition goal.

KEY TERMS

active rest
 annual training plan
 basic strength phase
 competitive period
 daily undulating periodization
 emphasis training models
 first transition period
 fitness–fatigue paradigm

general adaptation syndrome (GAS)
general preparatory phase
hypertrophy or strength–endurance phase
linear periodization
macrocycle
maintenance
mesocycle
microcycle
nonlinear periodization
parallel training models
peaking
periodization
preparatory period
recovery
restoration
second transition period
sequential training models
specific preparatory phase
stimulus–fatigue–recovery–adaptation theory
strength/power phase
supercompensation
traditional periodization

STUDY QUESTIONS

1. The duration of what phase of Selye's general adaptation syndrome will be lengthened if an athlete is under large amounts of psychoemotional stress?
 - a. alarm
 - b. resistance
 - c. supercompensation
 - d. exhaustion
2. The early stages of a long-term athlete development plan best aligns with which model of periodization?
 - a. parallel
 - b. sequential
 - c. emphasis
 - d. traditional
3. What can last between two and six weeks, with the most common duration used by strength and conditioning professionals being four weeks?
 - a. microcycle

- b. macrocycle
 - c. mesocycle
 - d. maxicycle
4. A preparatory period corresponds to what other part of the season?
- a. postseason
 - b. in-season
 - c. preseason
 - d. off-season
5. A hypertrophy phase of training best aligns with which phase?
- a. general preparatory
 - b. specific preparatory
 - c. precompetitive
 - d. main competitive